

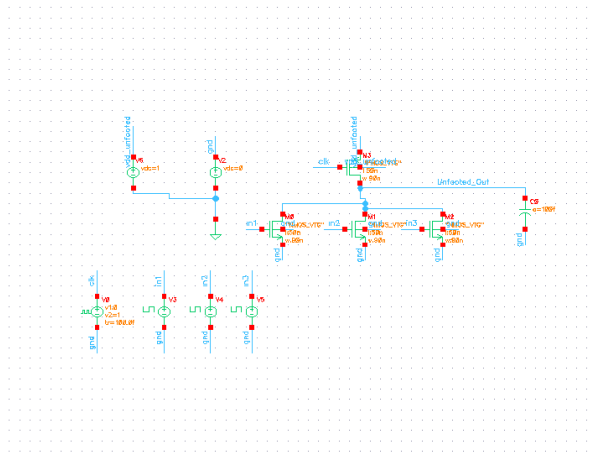
EC571 Lab5

Hoang Yen Tran

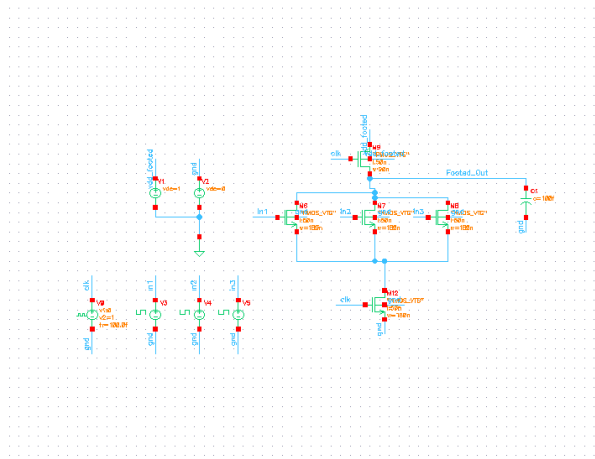
May 2026

1 Lab Exercise 1

1.1 Schematics



(a) Unfooted 3 Inputs NOR



(b) Footed 3 Inputs NOR

1.2 Logical Effort

Unfooted has the logical effort of $1/3$ and footed has logical effort of $2/3$.

1.3 Propagation delay

When measuring propagation delay, worst case propagation delay is when the output is pulled down by only 1 NMOS. This is because less current is conducted to discharge. In Figure 2, after a cycle of precharge, In1 and In2 is kept at 0, and In3 is driven to 1V during the evaluation, where clock is at VDD. Propagation delay is observed to be 535 ps for footed and 653 ps for unfooted. The faster discharge time is due to the footed case, the resistance of the in3 is less due to longer width, while the footed pulled to ground during evaluation. Therefore, footed logic discharge.

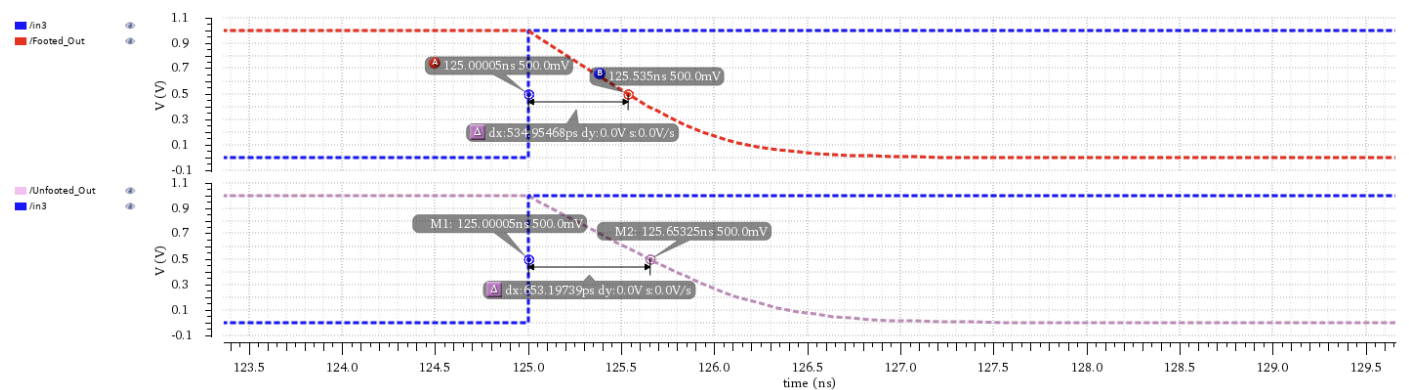
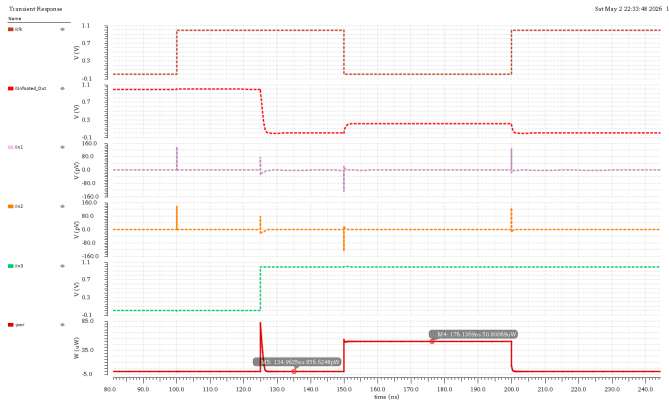


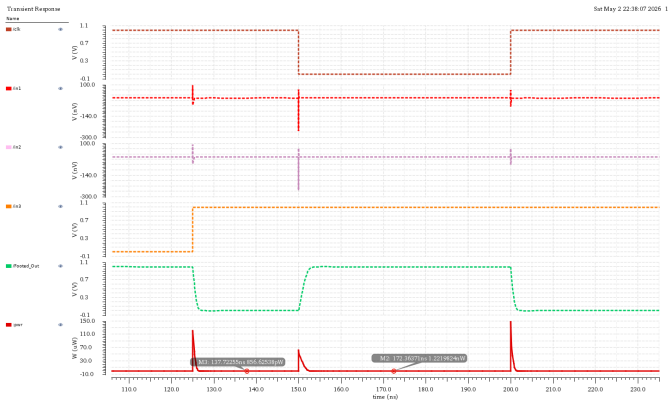
Figure 2: Propagation delay of unfooted and footed NOR gate

1.4 Static Power Consumption

Power consumption of unfooted 3 inputs NOR gate during during evaluation is both the same at 856.6 pW. However, during precharged, unfooted dynamic NOR logic consumes a more noticable amount of power of 50 uW comparing to that footed dynamic NOR logic with 1.221 nW consumption. Footed dynamic logic consumes less static power due to footed NMOS isolate the NOR logic from drawing power during precharged.



(a) Unfooted 3 Inputs NOR

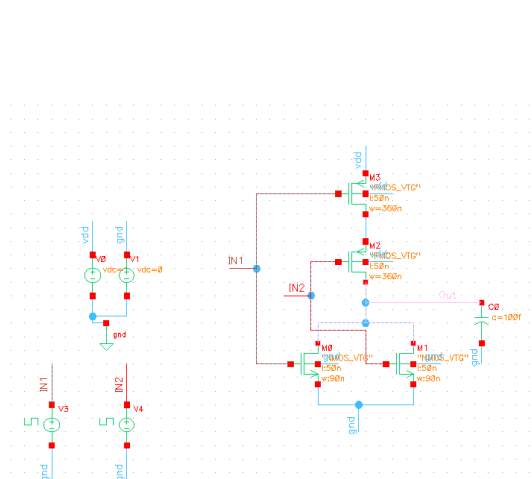


(b) Footed 3 Inputs NOR

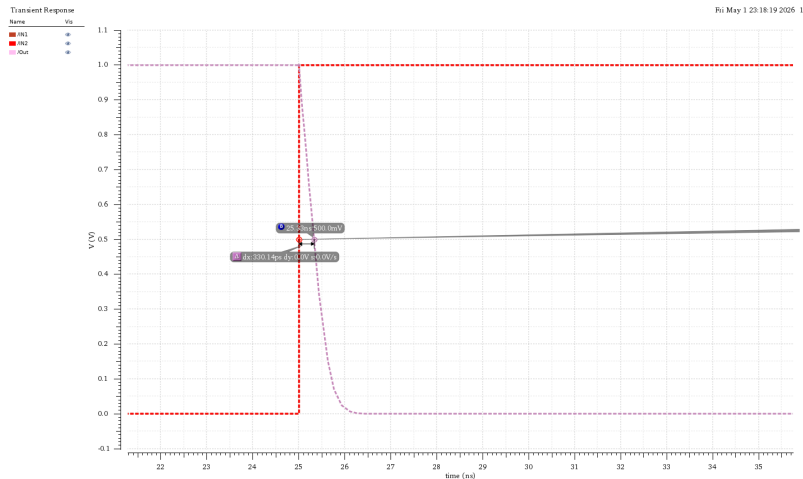
2 Lab Exercise 2

2.1 2-Input NOR gate with complementary CMOS logic

To observe the worst propagation delay, In1 is kept at 0, while In2 pulled to 1. Propagation delay is 330.14 ps. On



(a) CMOS 2 input NOR schematics



(b) CMOS's NOR Propagation delay

average, static and dynamic power consumption of CMOS NOR logic is 2.06 nW and 148.281 uW, respectively.



Figure 5: CMOS's NOR Power consumption

2.2 2-Input NOR gate with differential cascode voltage switch logic (DCVSL)

To observe the worst propagation delay, b is kept at 0, while a pulled to 1. Propagation delay is 49.27 ps.

On average, static and dynamic power consumption of DCVSL NOR logic is 3.544 nW and 112.7 uW, respectively.

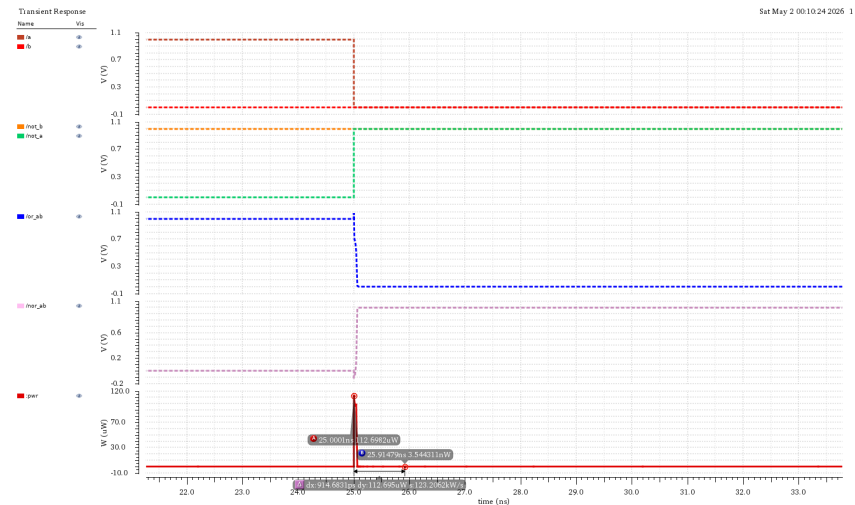
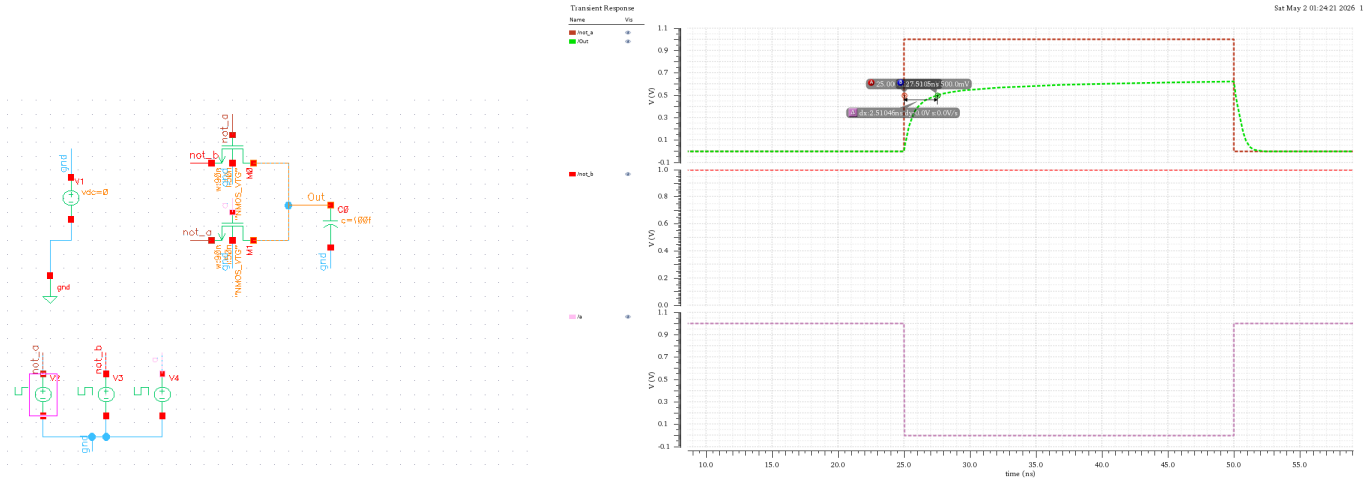


Figure 6: DCVSL's NOR Power consumption

2.3 2-Input NOR gate with pass transistor logic

Due to NMOS logic does not pass a strong one, output often reach about 0.7V. To observe the worst propagation delay, not_b is kept at 1, while not_a pulled up from 0 to 1. Propagation delay is 2.5 ps.



(a) Pass transistor logic's 2 input NOR schematics

(b) Pass transistor logic's NOR Propagation delay

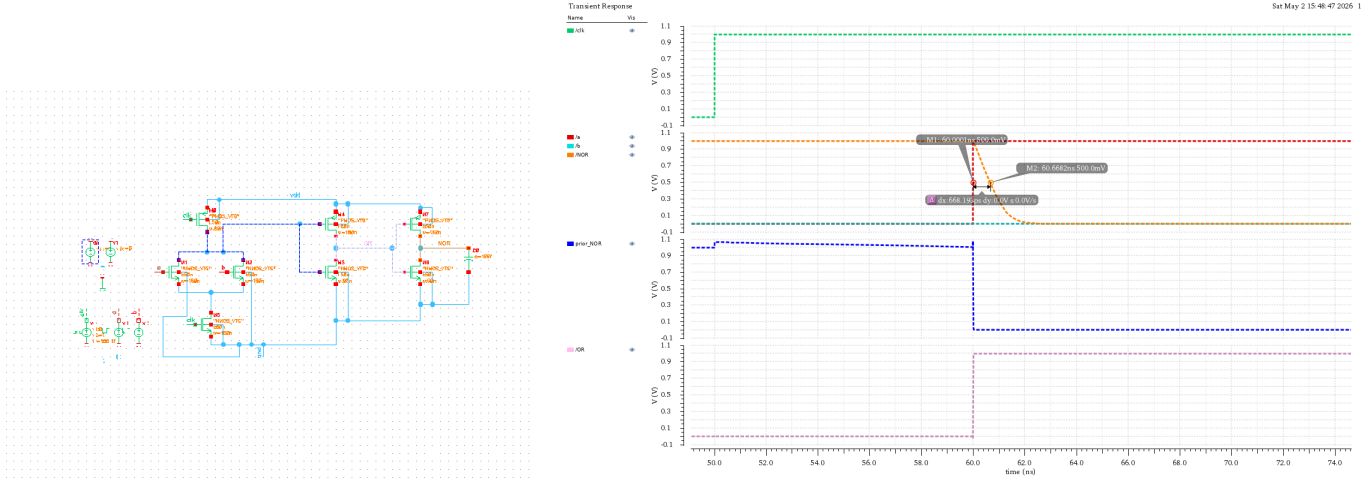
On average, static and dynamic power consumption of Pass transistor NOR logic is 163 nW and 81.7 uW, respectively.



Figure 8: Pass transistor logic's NOR Power consumption

2.4 2-Input NOR gate with dynamic domino logic

To observe the worst propagation delay, b is kept at 0, while a pulled up from 0 to 1 during evaluation. Propagation delay is 668.193 ps.



(a) Dynamic domino logic's 2 input NOR schematics

(b) Dynamic domino logic's NOR Propagation delay

On average, static and dynamic power consumption of Dynamic domino NOR logic is 3.5 nW and 123 uW, respectively.

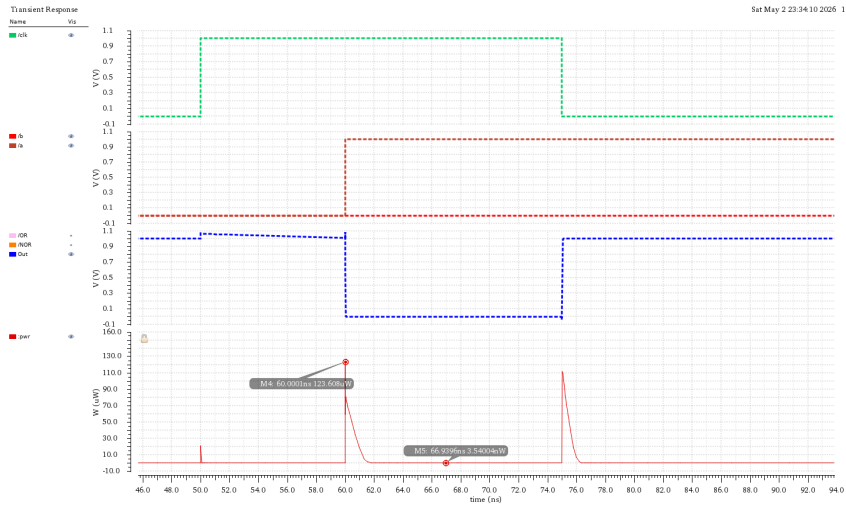


Figure 10: Dynamic domino logic's NOR Power consumption

2.5 Summary of all logic's Power consumption and delay

Overall, Pass transistor and DCVSL has the lowest delay. In terms of Static power consumption, pass transistor logic and DCVSL has the lowest consumption. In the case of dynamic power, CMOS and DCVSL has the lowest dynamic power during switching. Therefore, in implementing NOR, DCVSL has the most noticeable performance in terms of power and delay.